

ANSIBLE

1. Environment Setup

Option A: Local (Linux)

If you're using **Ubuntu/Debian**:

```
bash
```

```
sudo apt update
```

```
sudo apt install ansible -y
```

If you're using **RHEL/CentOS/Fedora**:

```
bash
```

```
sudo dnf install ansible -y
```

Verify installation:

```
bash
```

```
ansible --version
```

Option B: Cloud (e.g., AWS EC2 Ubuntu Instance)

1. Launch an EC2 instance (Ubuntu) in AWS.
2. SSH into it:

```
bash
```

```
ssh -i your-key.pem ubuntu@your-ec2-public-ip
```

3. Install Ansible on the EC2 instance using the same steps as above.

2. Basic Ansible Configuration

Edit or create an **inventory file**:

```
ini
```

```
# inventory.ini
```

```
[webservers]
```

```
localhost ansible_connection=local
```

3. Write a Basic Ansible Playbook

Create a YAML file called setup.yml:

```
yaml
```

```
---
```

```
- name: Basic setup
```

```
  hosts: webservers
```

```
  become: true
```

```
  tasks:
```

- name: Update apt cache

apt:

update_cache: yes

cache_valid_time: 3600

- name: Install Nginx

apt:

name: nginx

state: present

- name: Start and enable Nginx

service:

name: nginx

state: started

enabled: yes

4. Run the Playbook

Execute the playbook with:

```
bash
```

```
ansible-playbook -i inventory.ini setup.yml
```

This will:

- Update the package cache

- Install Nginx
 - Start and enable the Nginx service
-

5. Verify

After the playbook runs successfully:

- If you're on a local machine, go to <http://localhost> in your browser.
- If on a cloud instance, visit <http://your-ec2-public-ip> to see the Nginx welcome page.